Which hand?

Detecting left- vs right-hand nose touches from smartphone motion sensors

Axel Fridman

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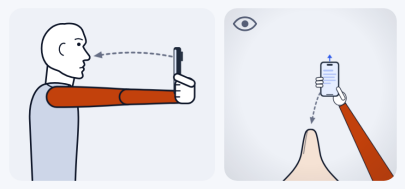
Abstract

A person brings a smartphone from arm's length to the nose with the left or the right hand. From the phone's motion sensors (a web browser, about 60 Hz), this report predicts which hand as a calibrated probability $P(\text{right})$, for persons the model has never seen. The data are 598 instructed trials of 20 persons, and every evaluation leaves one person out. The phone's early rotation about gravity changes sign with the hand. A pre-specified rule chose a one-dimensional convolutional network among 17 candidates; the nested, selection-aware estimate of this selection, with the candidates fixed, is an accuracy of 88.5 % [84.1, 92.3] (area under the ROC curve, AUC, 0.945). With its in-fold Platt calibrator the network's probabilities are consistent with calibration (expected calibration error, ECE, 0.026, null band up to 0.042; slope 0.97 [0.69, 1.50]), and three attempts raise its accuracy to 96.4 % (exploratory). Without a start cue, a simpler reference classifier, not the final model, shows a hand for 76.4 % of the recordings and is right on 79.9 % of them (forced choice 74.6 %; exploratory, the best of 22 pipelines). The main gaps are 2 left-handed persons, right-hand free-choice trials only and no cue-free run of the final model yet.

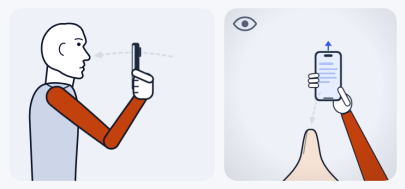
Key results	
Unseen persons	Nested, selection-aware accuracy 88.5 % [84.1, 92.3], AUC 0.945 [0.906, 0.975] (candidates fixed). Section 5.
The percentage	With its in-fold Platt calibrator, consistent with calibration: ECE 0.026 (null band up to 0.042), slope 0.97 [0.69, 1.50], intercept +0.07 [−0.25, +0.39]; no further remedy helps.
Three attempts	Accuracy of the mean $P(\text{right})$ 96.4 % [93.2, 98.9] (exploratory); the displayed aggregate is neither over- nor under-confident on average (calibration in the large only).
Natural use	Right-hand recall 85.0 % (17 of 20); all free-choice trials are right-hand, so a constant right answer would score 100 %.
No start cue	The production simulation's (lane m4) reference classifier yaw8, *not* the final model (exploratory, best of 22 pipelines): a hand shown for 76.4 % of the recordings, right on 79.9 % of those; forced choice 74.6 %; 72.8 % [68.9, 76.6] with half credit for abstentions. Section 6.

1 Introduction

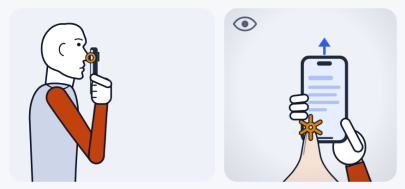
The question. A person holds a smartphone at arm's length and brings it to the nose with the left or the right hand (Figure 1). The task, set by the principal investigator (PI), is to



(a) Arm extended, still



(b) The approach



(c) Nose touch and hold

Figure 1: The gesture as the app shows it (right hand, upright grip, natural speed). The model reads the approach.

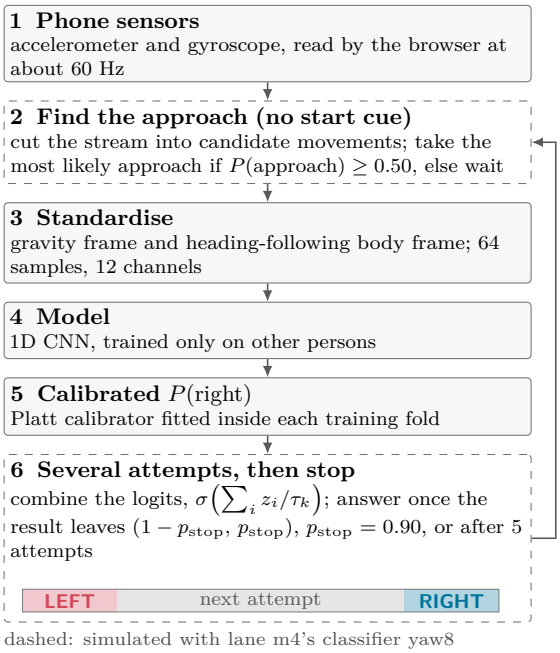


Figure 2: From sensors to an answer. Steps 3–5 are evaluated on the stored approach windows by leaving one person out; the dashed cue-free steps were simulated with lane m4's reference classifier, not yet with the final model.

predict which hand from the motion sensors that the browser's DeviceMotion interface delivers at about 60 Hz [1], as a calibrated probability $P(\text{right})$ that the right hand moved the phone (of the touches given 80 %, about 80 % should be right-hand touches), for persons the model has never seen.

Why it is hard. Persons, phones and grips differ, and the device axes turn with the grip. The two hands make mirror-image movements, so most simple descriptors (speed, duration, path length) are the same for both, and in production there is no start cue: the page must find the approach in a continuous stream.

The approach. The left–right mirror symmetry of the body shows which quantities can carry the hand (Section 3). Each trial is standardised into a gravity frame and a heading-following body frame, four model families are compared by leave-one-person-out (LOSO) cross-validation, a pre-specified rule picks the final model, and a simulation then removes the start cue (Figure 2).

Status. This preliminary version shortens the full report (report/main.pdf in `https://github.com/AxelFridman/nosetouch-study`), which states every decision with its evidence. The analysis ran in lanes: the three model-family lanes (m1–m3), preprocessing (p1) and the

production simulation (m4) each passed two independent audits; the final-model lane is a first pass, not yet independently audited, and the final model has not yet run in the cue-free simulation.

Related work. Most prior studies infer the hand from touch traces or grip during ordinary phone use, and several split the data by trial or at random, which overstates the accuracy for new persons [2–6]. Opposite rotation senses of the two hands were seen with wrist sensors in a clinical finger-to-nose test, without a classifier [7], and modern classifiers are often over-confident [8], so this study evaluates person-wise and checks calibration.

2 Data

App and protocol. The data were recorded with NoseTouch, a web app that runs in the phone's browser and guides the participant hands-free with spoken and visual prompts. Tier 0 of a visit is one free-choice trial, whose hand the participant picks and reports afterwards (the natural-use set, scored but never trained on); tiers 1–4 prompt the hand, the grip and the speed in blocks of one hand (full report, Table 2), and the median visit took 9.7 min.

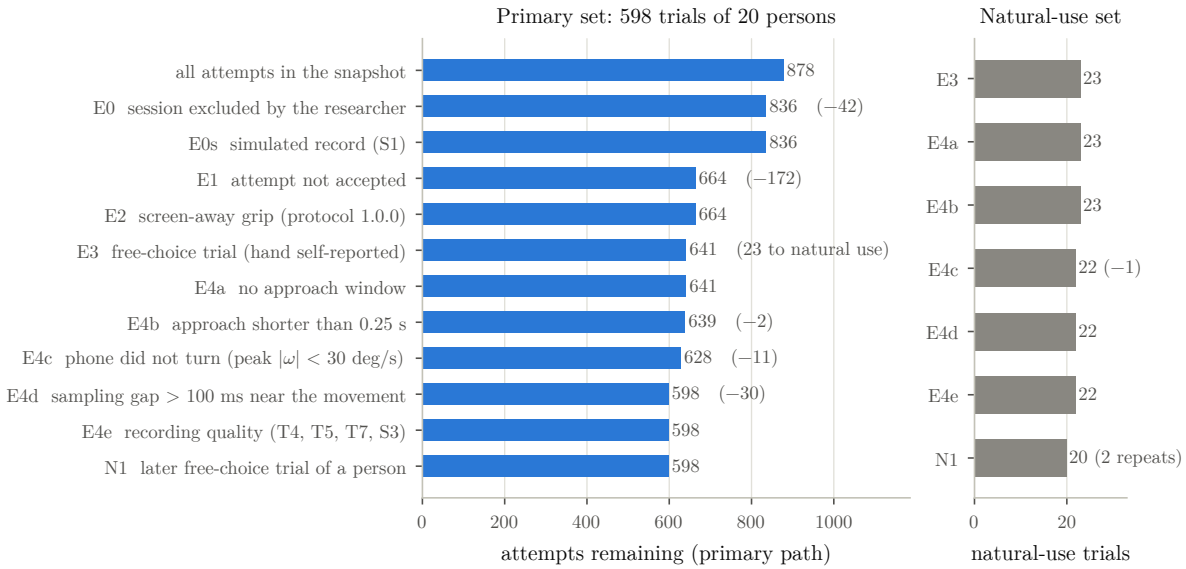


Figure 3: The exclusion cascade, applied in order; an attempt is charged to the first rule it fails. It yields the primary set for training and evaluation and the natural-use set, which is only scored.

Participants. The participants were Bar-Ilan University students, most of them master's students, each offered sweets for their time; a video shows a data-collection session[1]. One additional participant, PD01, was analysed separately in an exploratory case study and never modelled (Section 7).

[1] A video of a data-collection session: https://drive.google.com/file/d/1V9PF623HkJ8MjR6loJhUHnDkCx60W-Jl/view?usp=sharing.

Curation. The data snapshot holds 878 attempts, and a pre-specified exclusion cascade keeps 598 primary trials and 20 natural-use trials (Figure 3). Quality flags are kept and scored apart rather than excluded, so the models see the data production will see, and labels are never changed.

The dataset. The 20 persons are 12 women and 8 men aged 19 to 29 years, 2 of them left-handed; 12 recorded on Android and 8 on iOS. The primary set has 302 left-hand and 296 right-hand trials.

Gaps. With 2 left-handed persons no effect of handedness can be estimated, and with 8 iPhone users platform comparisons have little power. All 20 free-choice trials are right-hand trials, so natural use measures right-hand recall only: a constant right answer would score 100 %.

3 Why the hand is detectable

The mirror. Let $\mathbf{f}$ be the specific force (the browser's acceleration including gravity), $\mathbf{a}$ the acceleration and $\boldsymbol{\omega}$ the angular velocity, all in device coordinates. Mirroring the body across its sagittal plane and relabelling the device's x axis gives, with $\mathbf{S} = \mathrm{diag}(-1, 1, 1)$, for every grip and every movement

$$\mathbf{f}' = \mathbf{S}\mathbf{f}, \qquad \mathbf{a}' = \mathbf{S}\mathbf{a}, \qquad \boldsymbol{\omega}' = -\mathbf{S}\boldsymbol{\omega}. \tag{1}$$

Chiral quantities. Angular velocity is a pseudovector, so with $\hat{\mathbf{g}}$ the unit vector pointing up, the yaw rate about gravity, $\boldsymbol{\omega} \cdot \hat{\mathbf{g}}$, changes sign under the mirror and is unchanged by any rotation of the phone in the hand. Only such chiral (mirror-odd) quantities can carry the hand: a mirror-even quantity has the same distribution for both hands whenever the two hands move as mirror images.

The mechanism. Early in the approach the forearm swings toward the body's midline, so, seen from above, the phone turns counter-clockwise in the right hand and clockwise in the left. Near the nose the wrist turns the phone partly back, so the early turn carries the clearest signal.

The evidence. The yaw about gravity over the first half of the approach separates the hands with a pooled area under the ROC curve (AUC) of 0.85, against 0.77 for the net yaw over the whole approach, a difference not tested in a paired comparison (Figure 4). Its within-person AUC is above 0.5 for 20 of 20 persons, its sign alone, with no training, classifies 79 % of the trials, and it separates the hands with the same sign on both platforms (AUC 0.86 on Android, 0.84 on iOS).

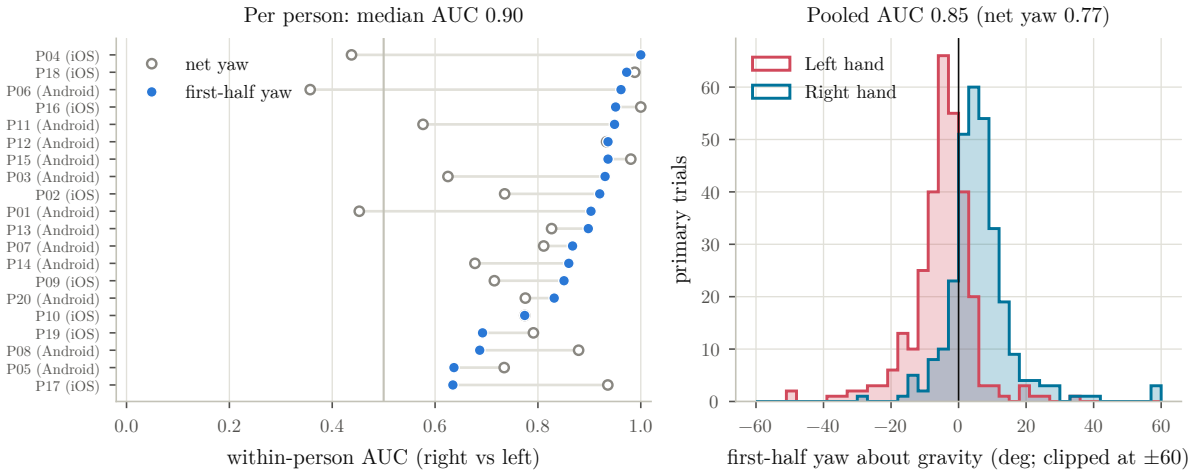


Figure 4: The yaw about gravity over the first half of the approach: AUC within each person (left) and the pooled distributions (right). Its pooled AUC is higher than that of the net yaw (no paired test), so the models get time-resolved inputs.

Chiral against mirror-even features. In the feature catalogue of the interpretable models, chiral features reach a LOSO AUC of up to 0.87, while every mirror-even feature stays between 0.46 and 0.53 (full report, Appendix C.2). The two hands are close to, but not exactly, mirror images, so mirror augmentation (a mirrored trial counted as a trial of the other hand) is a plausible, not an exact, way to share data between them.

Consequences. The models therefore get time-resolved chiral inputs, a body frame whose chiral channels do not depend on the grip, and mirror augmentation. The return of the arm is a time-reversed approach: on the 288 trials with a clean return window, the first-half yaw rule is right on 76 % of the approaches but on only 17 % of the returns, so a cue-free detector that scores the return answers the wrong hand.

4 Methods

Standardisation. Every trial is resampled onto a uniform 60 Hz grid, the iOS accelerometer sign is normalised, and a gravity frame is built without the magnetometer by carrying the gravity direction at rest through the movement with the gyroscope. A body frame follows the phone's heading, so a rotation of the phone in the hand leaves its channels unchanged; all channels are low-pass filtered at 15 Hz, and the approach window stored by the app (the oracle window) is time-normalised to 64 samples.

Data treatment. In the preprocessing lane p1, a fixed panel of pre-specified models judged each data-treatment choice against its alternatives, and all 11 defaults were kept (full report, Appendix B).

Evaluation. Every evaluation is LOSO: a model is trained on all other persons and tested on the held-out person, once for each of the 20 persons, with every hyperparameter, calibrator and aggregation scale fitted on the training persons only. All 95 % confidence intervals (CIs) are person-cluster bootstrap intervals over 2000 resamples of persons [9]; paired comparisons use an exact sign-flip test on the per-person differences, p-values are adjusted within a named family by Holm's method or, where stated, by the Benjamini–Hochberg method, and single pre-specified tests are reported unadjusted.

Selection. A configuration chosen by its LOSO result among several is exploratory, and its number of configurations tried is reported with it. The headline of a selection is its nested estimate: the whole selection is repeated inside each outer fold on the training persons only, so the held-out person never influences the choice.

Abstention and natural use. A pipeline may abstain by returning $P(\text{right}) = 0.5$ exactly; an abstention is never a right-hand answer and earns half credit, and for such a pipeline the coverage (the share answered), the accuracy on the answered trials and the forced-choice accuracy are reported. Natural use is scored as right-hand recall on the free-choice trials; all are right-hand trials so far, so a constant right answer would score 100 %.

Metrics. Accuracy reads $P(\text{right})$ above 0.5 as right; the AUC and the log loss measure discrimination and the quality of the probabilities. Calibration is judged by three numbers together: the expected calibration error (ECE) against its perfect-calibration null band (the ECE values a perfectly calibrated model shows by chance), and the slope and intercept of a logistic recalibration, near 1 and 0 for a calibrated model.

Several attempts. Attempts $i = 1, \ldots, k$ of one person and hand are combined as

$$P(\text{right})_k = \sigma\left(\frac{1}{\tau_k}\sum_{i=1}^{k} z_i\right), \tag{2}$$

where z_i is the logit of attempt i, σ the logistic function and τ_k a scale fitted inside each fold; the k-attempt results use the 19 persons with enough trials of each hand.

Model families. Four families were developed in parallel:

- rules and interpretable models (lane m1): the sign of the first-half yaw and antisymmetric logistic models on six physics features (PHYS6);
- classical machine learning on 691 engineered features (lane m2): logistic regression, support vector machines, LightGBM and stacks;
- time-frequency and deep sequence models (lane m3): MiniRocket [10] and one-dimensional convolutional networks (1D CNN);
- the reference floor yaw8: a mirror-augmented logistic regression on the cumulative yaw at 8 instants, fixed before any model was built.

The cross-lane rule. With the finalists of lanes m1–m3 and p1, the reference floor and a few cross-lane variants, 17 candidates entered a selection whose rule was fixed in advance: find the lowest log loss in production form and the leaders that paired comparisons cannot separate from it (Holm), drop the leaders with a Holm-significant accuracy drop on an unseen grip, a natural-use shortfall or a screen-away failure that the rest-pose check cannot repair, and take the most portable of the rest. Rerun inside every outer fold, the same rule gives the nested, selection-aware estimate; it treats the candidates as fixed and does not repeat the lanes' own choices made on the outer folds.

5 Results

Model comparison. Most candidates beat the reference floor yaw8 (77.9 % [73.1, 82.9]): 11 of 16 are more accurate after Holm adjustment over these comparisons (Figure 5). The best differ little: the candidate with the lowest log loss and those that paired comparisons cannot separate from it (Holm over its 16 comparisons) form 6 leaders, and since 7 candidates were chosen on the outer folds in their own lane, the comparison is exploratory.

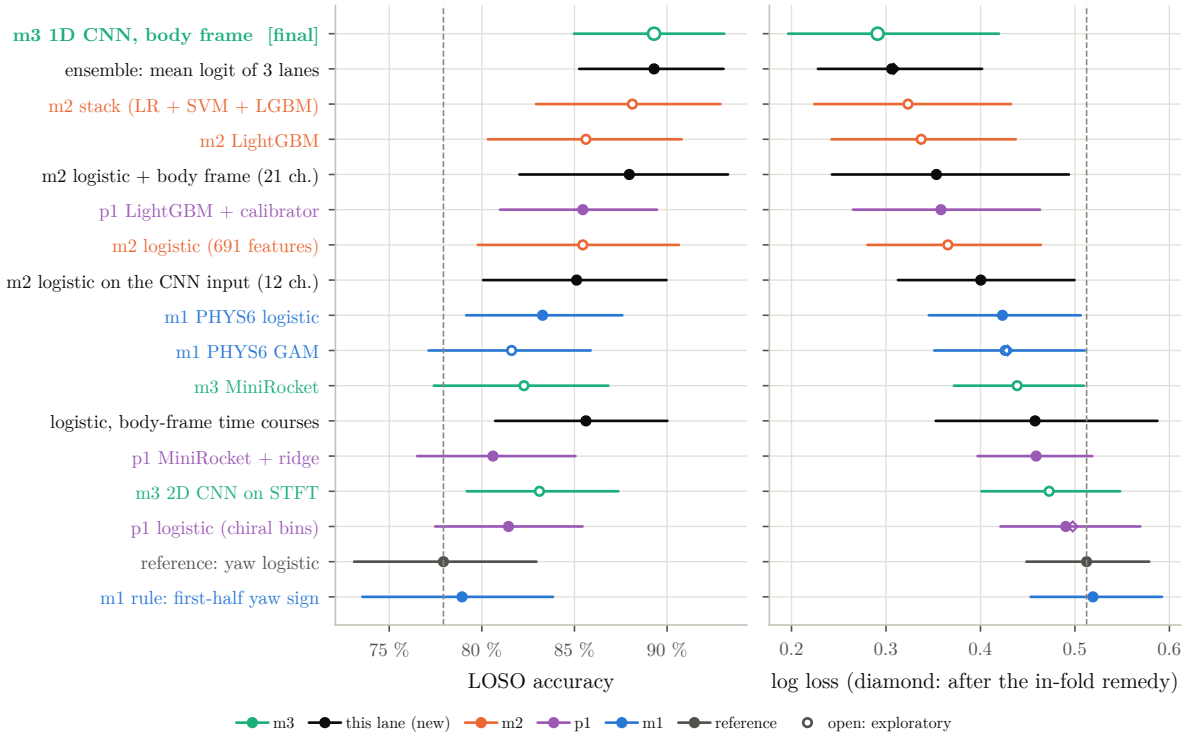


Figure 5: The 17 candidates as their lanes deliver them (LOSO accuracy and log loss; dashed: the reference floor; this lane: the final-model lane). The forest is exploratory; the nested estimate is the headline.

The final model and its selection-aware estimate. The final model is the body-frame 1D CNN, a convolutional network on 12 gravity- and body-frame channels with the lowest log loss in production form. The nested, selection-aware estimate, with the candidates fixed, is an

accuracy of 88.5 % [84.1, 92.3], an AUC of 0.945 [0.906, 0.975] and a log loss of 0.307 [0.212, 0.432]; inside the outer folds the selection chose the CNN in 16 of 20 (Figure 6).

The shipped model's own numbers. The shipped model's LOSO accuracy, 89.3 % [85.0, 93.1], is exploratory: it was the best of 16 configurations in its lane, then chosen among 17 candidates, an optimism of +0.8 [−0.2, +2.3] points over the nested estimate. Its body-frame input was also designed after outer-fold results had been seen, an optimism no nested estimate removes, and the rest of this section, which describes the shipped model, shares this status.

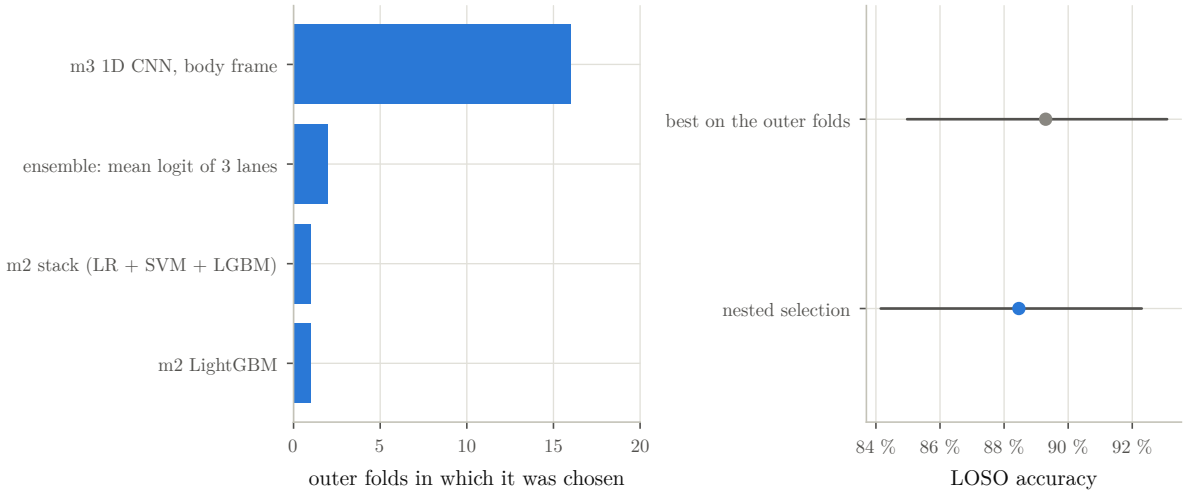


Figure 6: The nested cross-lane selection: how often each candidate was chosen over the 20 outer folds (left), and the selection-aware estimate against the naive best pick (right).

Robustness to an unseen grip. Trained on upright grips only and tested on the sideways grips, the leaders that read device-frame channels change by −19.8 to −31.0 points (each Holm $p \leq 0.014$ over the 17 grip tests; Figure 7). The body-frame CNN changes by −4.1 [−7.0, −1.4] points (unadjusted $p = 0.016$; Holm $p = 0.188$ over the same tests), and of the two leaders without a Holm-significant drop the rule took the CNN as the more portable.

Screen facing away. A phone held with the screen away defeats every model that reads the screen-relative frame: on synthesised screen-away twins of the recorded trials the CNN falls to 28.6 %. A model-free check of the rest pose, exported with the model, and an un-flip recover 86.6 %.

Calibration. With its in-fold Platt calibrator [11], the final model's probabilities are consistent with calibration (Figure 8): ECE 0.026, inside its null band (up to 0.042), slope 0.97 [0.69, 1.50] and intercept +0.07 [−0.25, +0.39]. No further remedy helps: Platt, temperature, isotonic or beta calibration on top each raise the log loss slightly, none detectably (Holm $p = 1.000$ for each, over the four remedies).

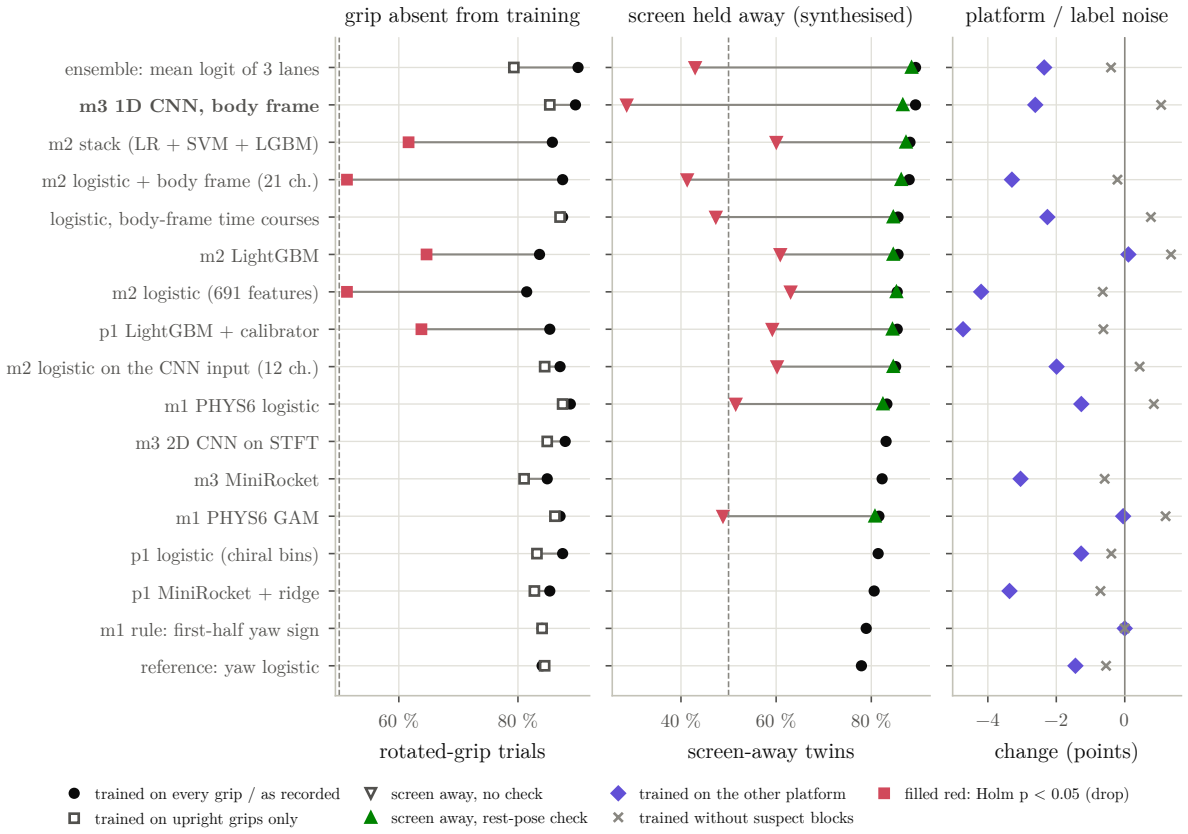


Figure 7: Robustness of every candidate (bold: the final model) to an unseen grip (left), a screen-away hold (middle), and training on the other platform only or without suspect label blocks (right). Device-frame models fail an unseen grip, which removes them from the selection.

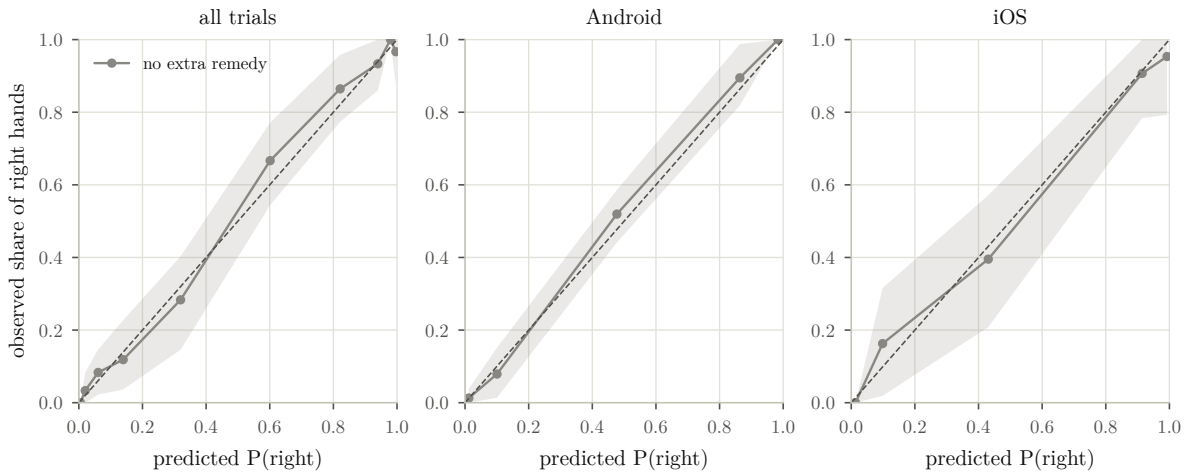


Figure 8: Reliability of the final model with its in-fold Platt calibrator, overall and per platform (person-bootstrap 95 % bands). The percentage shown is consistent with calibration.

Several attempts. Combining attempts helps (Figure 9): the mean $P(\text{right})$ of $k = 3$ attempts of one person and hand reaches an accuracy of 96.4 % [93.2, 98.9]. The displayed aggregate of equation (2) is right in 96.3 % and neither over- nor under-confident on average (mean confidence minus accuracy −0.008 [−0.024, +0.016]), a check of calibration in the large only.

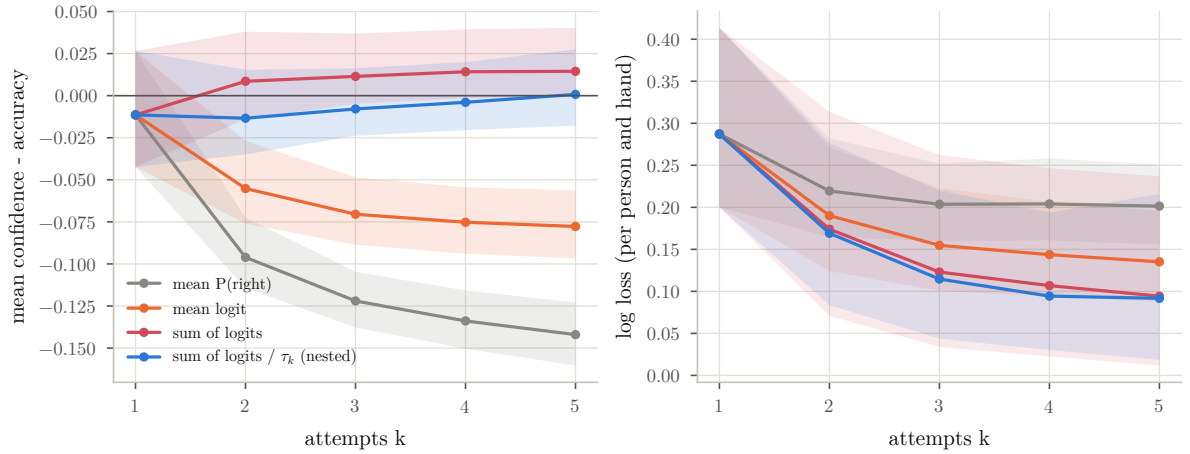


Figure 9: Several attempts of one person and hand: over-confidence (left) and log loss (right) of four aggregates. The displayed scaled sum of logits (equation 2) is neither over- nor under-confident on average.

Who and where it fails. The accuracy is 90.6 % on Android and 87.0 % on iOS (not compared by a test) and below 80 % for 3 persons; the two left-handed persons score 96.4 % and 88.0 % (Figure 10). Of the 64 errors, 24 are confident (a probability below 0.2 for the true hand), and they cluster in a few persons (7 in P09, 5 in P20).

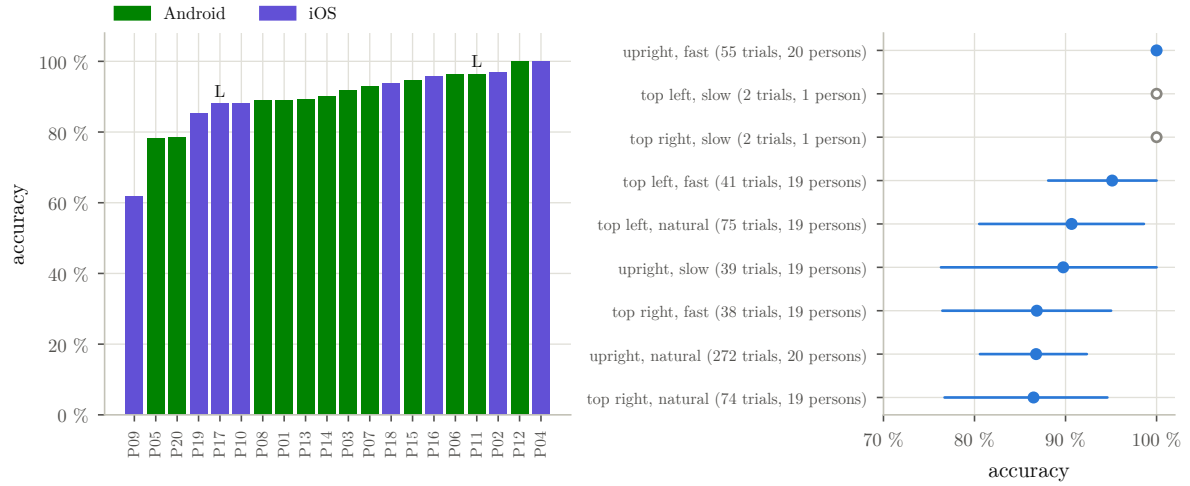


Figure 10: The final model by person (left; L: the two left-handed persons) and by grip × speed cell (right; CIs only for cells with enough persons). 3 persons fall below 80 %.

Natural use. On the free-choice trials the final model's right-hand recall is 85.0 % (17 of 20), with no detectable shortfall against the same persons' instructed trials (Holm $p = 1.000$ over the 17 candidates). All free-choice trials are right-hand trials, so a constant right answer would score 100 % and this recall says nothing about the left hand.

More persons? An inverse power law fitted to random subsets of training persons extrapolates the accuracy to 89.6 % [88.8, 90.8] at 40 persons, a band that understates the uncertainty. By this extrapolation more persons of the same kind would add little average accuracy, so new persons are needed above all for left-handers and left-hand free choice (full report, Appendix F).

6 Towards production: no start cue

Scope. In production the page receives a continuous stream with no start cue: it must find the approach, reject other movements, decide when to answer and show a calibrated P(right). Lane m4, the production simulation, ran this pipeline on the full recordings with its reference classifier yaw8, *not* the final model, to measure what detection and stopping cost, so every cue-free number in this section is yaw8's; running the final model inside it is the next step.

Finding the approach. The baseline detector v0 takes the first movement after a fixed stillness: its first detection is the approach in 53 % of the recordings and an earlier movement in 28 % (Figure 11). The production detector scores candidate movements with a segment classifier that does not use the hand and accepts the most likely one when its probability of being an approach is at least 0.50: its scored segment is the approach in 67 % of the recordings and an earlier movement in 5 %, and it rejects 17 %.

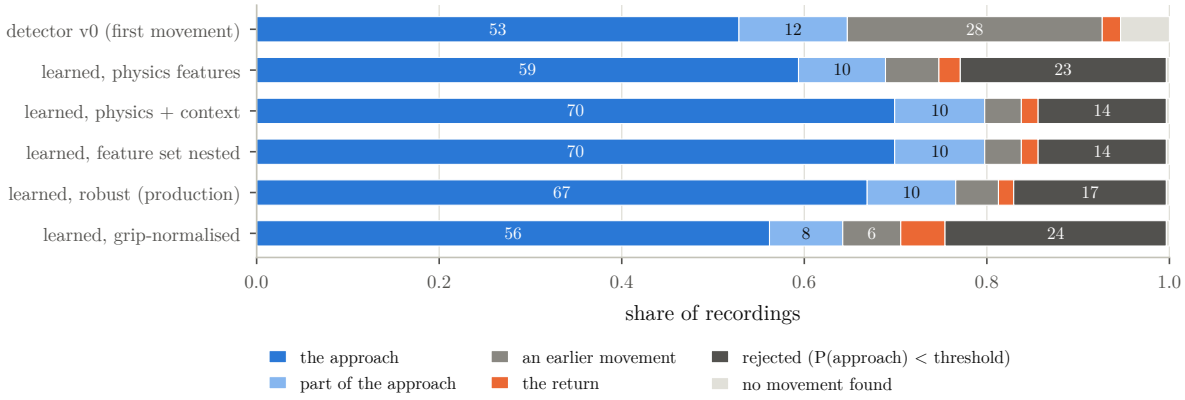


Figure 11: What each cue-free detector hands to the hand classifier on the 598 full recordings (percentages). The production detector scores far fewer earlier movements than the baseline v0; it rejects 17 % of the recordings.

What detection costs. The classifier yaw8 reaches 77.9 % [73.1, 82.9] on the stored windows and 72.8 % [68.9, 76.6] in the production pipeline, where a rejection or an output of exactly 0.5 is an abstention with half credit. The pipeline shows a hand for 76.4 % of the recordings,

right on 79.9 % of those (forced choice 74.6 %); detection changes the accuracy by −5.1 [−7.6, −2.7] points (Holm $p = 0.047$, family of 26 comparisons); the pipeline, best of 22, is exploratory.

Stillness and screen away. The detector relies on the stillness the app enforced: the pipeline falls from 73.1 % with 1 s of stillness before the approach to 56.4 % with none, when it answers only 33 % of the recordings. A model-free check of the rest pose stops 95.7 % of synthesised screen-away twins and 4.3 % of the recordings held toward the person.

Several attempts and the stopping rule. On synthetic sessions spliced from each person's recordings, combining up to three accepted attempts gives 76.5 % [69.2, 83.7], against 88.4 % [82.0, 94.5] with the true approaches only, because only 77 % of the accepted attempts are approaches (Figure 12). The stopping rule answers once the aggregated $P(\text{right})$ leaves the band between $1 - p_{\text{stop}}$ and $p_{\text{stop}} = 0.90$, or after 5 attempts, and reaches 83.2 % [74.6, 90.8] after 3.66 attempts on average; its scale was added after the held-out sessions had been seen, so it is post hoc and exploratory (6 configurations).

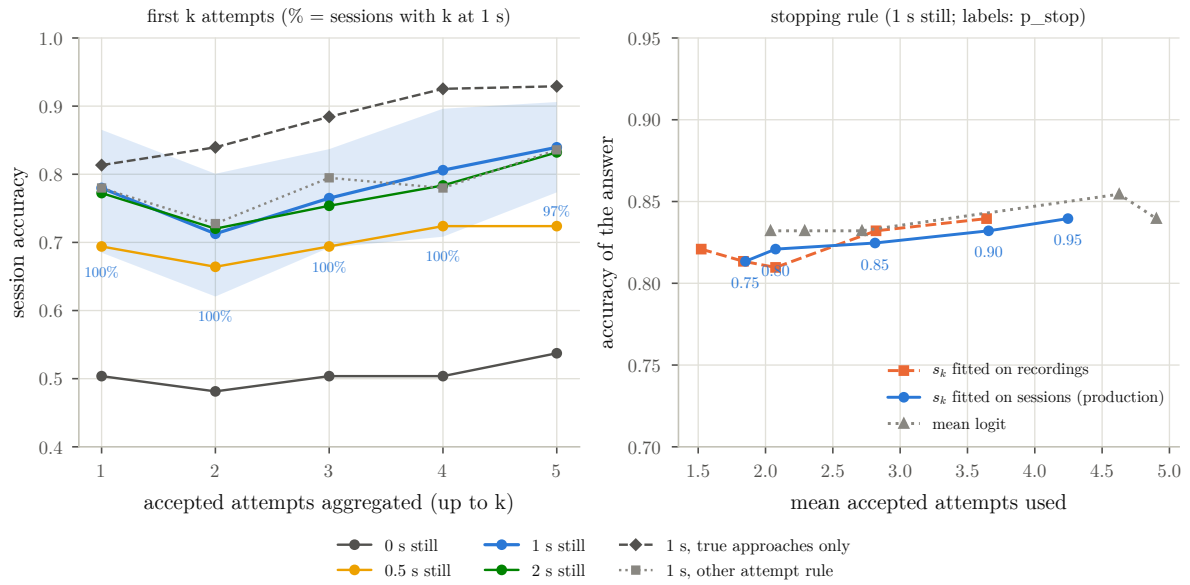


Figure 12: Synthetic multi-attempt sessions with yaw8. Left: accuracy after the first k accepted attempts; right: the stopping rule's accuracy against the attempts it uses, for several p_{stop} ($s_k = k/\tau_k$ scales the mean logit, equation (2)).

The calibrated output. For one touch, the cue-free output is consistent with calibration: ECE 0.048 (null band 0.020 to 0.060), slope 0.94 [0.75, 1.18] and intercept −0.13 [−0.29, 0.03]. At the stop the ECE of 0.107 lies at the upper edge of its null band (up to 0.108), a borderline result that needs a prospective test.

Natural use: the main failure mode. The detector rejects 9 of 20 free-choice recordings (45 %, against 17 % on the primary set; Poisson-binomial $p = 0.0015$, unadjusted, a single pre-planned

test). A hand is shown for 10 of the 20 (1 accepted recording got exactly 0.5), with a right-hand recall of 90.0 % (forced choice 72 %); all are right-hand trials, so a constant right answer would score 100 %.

For the user and the page. Three instructions follow: hold the phone with the screen facing you, tilted slightly up; keep still for 1 s before the movement; and make several attempts. The decision comes a median 754 ms after the screen first touches the nose; the exported final model (46084 fitted numbers, 1585 kB) reproduces the fitted pipeline's $P(\text{right})$ to within 2.2e-16 but has not yet run behind the cue-free detector (full report, Appendices G.7 and F.14).

7 Secondary analyses

The same recordings carry more than the hand. These models feed nothing the page shows, and all use LOSO with balanced accuracy, the mean recall over the classes.

Instructed speed. On label-blind windows, a pre-specified multinomial logistic model on mirror-invariant kinematic features recognises the instructed speed at a balanced accuracy of 50.1 % [42.9, 56.4] (chance 33.3 %). The protocol confounds speed with grip, however: the grip alone predicts the speed at 48.0 % [43.8, 52.9], and the model does not beat it (Holm $p = 1.000$ over the balanced-accuracy comparisons of the speed models).

Grip. The grip is recognised at 95.2 % [89.7, 100.0] from the up vector while the phone rests before the movement. Its 18 errors come from 3 persons and were probably trials held in another grip than instructed, although no independent record confirms it.

Dominant hand. Mirror-invariant kinematics show no detectable dominant-hand signal: the pooled AUC is 0.491 [0.453, 0.533], against a within-person permutation null with mean 0.495 ($p = 0.582$, unadjusted, a single pre-specified test; Figure 13).

Demographics. Production has no demographics. In a quick, unaudited first pass, giving a second-stage model on the final model's logit the persons' age, sex, height, handedness and usual phone hand changes the accuracy by −3.2 [−6.3, −0.7] points (Holm $p = 0.062$ over the two add-on variants), so they are not added.

An exploratory case study: PD01. One additional participant with Parkinson's disease, PD01, was compared with the 20 modelling persons in a descriptive case study; no model was trained, tuned or scored on PD01's trials. PD01 lies beyond every reference person in 9 of 42 pre-specified movement characteristics, but 2 reference persons, judged on as many trials, reach that count, and no difference survives the Benjamini–Hochberg adjustment over the pre-specified characteristics. PD01 is unusual, not unique, and nothing here is a diagnostic claim (full report, Appendix I).

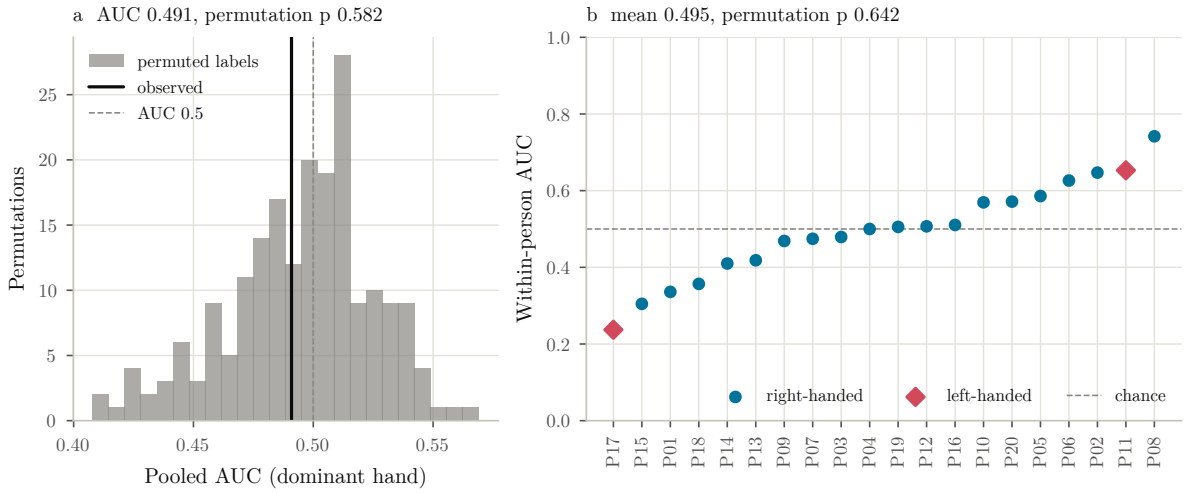


Figure 13: Mirror-invariant kinematics show no detectable dominant-hand signal: the pooled AUC against its within-person permutation null (a) and the AUC within each person (b).

8 Discussion

What the results mean. Early in the approach the phone turns about gravity in opposite senses for the two hands, so the hand can be read from the motion sensors for persons the model has never seen: the cross-lane selection that chose the body-frame 1D CNN has a selection-aware accuracy of 88.5 % [84.1, 92.3], the CNN's body frame limits the change on an unseen grip to −4.1 points (device-frame leaders: −19.8 to −31.0), and with its in-fold Platt calibrator its percentage is consistent with calibration. The main risk lies before the classifier: finding the approach without a cue costs accuracy, and free-choice movements are often rejected.

Limitations.

- Only 20 persons, mostly students; 2 left-handed and 8 on iOS.
- One free-choice trial per person, all right-hand: a constant right answer would score 100 %.
- The cue-free numbers come from lane m4's yaw8 classifier, not the final model.
- Screen-away holds and multi-attempt sessions are synthesised, not recorded.
- The shipped model's own numbers are optimistic: it was the best of 16 configurations, then of 17 candidates, and its body-frame input was designed after outer-fold results had been seen. The nested estimate removes only the optimism of choosing among the 17 candidates, not that of the lanes' own outer-fold choices or of the post hoc body-frame design.

Next steps.

1. Run the final model in the cue-free simulation; teach the detector free-choice approaches.
2. Build the inference page from the exported model, with the rest-pose check, the stillness check and the displayed aggregate of equation (2).
3. Record more persons, especially left-handed persons, and left-hand free-choice trials.
4. Let the PI evaluate the page prospectively on new persons.

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